Sudhiksha M K

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Profile

Python focused Software Developer with industry experience in backend engineering, data modeling, database management, and workflow automation. Proficient in Python, SQL, PostgreSQL, and building structured data processing solutions. Experienced in developing automated reporting workflows, data pipelines, and scalable backend services. Interested in data engineering, large-scale data extraction, and distributed data processing systems.

Professional Experience

Consultant Software Developer (Part-Time), VISTA Engg Solutions 11/2025 – 04/2026

Software Developer Intern, VISTA Engg Solutions 05/2025 – 11/2025

- Designed relational data models and role-based data access structures for an enterprise workforce management system, organizing task assignments and hierarchical organizational data.
- Developed a QA automation and reporting platform that collected test outcomes into structured datasets and generated automated status reports, reducing testing and reporting turnaround by 50%.
- Implemented template-driven data processing and automated report generation, reducing manual reporting effort by 60% and improving data consistency.
- Implemented document version tracking, change management, and digital signature verification to maintain data integrity and traceability.
- Developed backend services and database integrations using Python, Flask, Node.js, and PostgreSQL to support persistent data management and enterprise workflows.

Education

M.Sc. Software Systems (Integrated), Coimbatore Institute of Technology 2022 – Present
CGPA-8.38 (Till semester-8)

Skills

Languages & Tooling — Python, C++, C, JavaScript, HTML/CSS, Git
Database and Data Management — PostgreSQL, MySQL, MongoDB, Supabase, Neon DB, data modeling, schema design, RBAC
Data Engineering & Analytics — Pandas, Data Preprocessing, Data Cleaning, ETL Concepts, Data Transformation, Data Validation, Report Automation
Backend & APIs — Node.js, Flask, FastAPI, REST APIs, Express, PHP
Data Visualization & Application Layer — React, Vue.js, dashboards & interactive UI for data-driven decision-making, Figma

Projects

FoodFlow IQ – Food Delivery Analytics & Predictive Intelligence System , *(Python, FastAPI, Pandas, PostgreSQL, Scikit-learn)*

- Developed a dynamic CSV ingestion and ETL pipeline with automated schema detection, data profiling, validation, and preprocessing.
- Implemented reusable Pandas workflows for missing-value handling, duplicate removal, data transformation, and derived feature generation.
- Designed a hybrid storage architecture using SQLAlchemy, PostgreSQL, and Parquet for dataset metadata and persistent data storage.
- Built analytical and predictive modules for sales aggregation, RFM segmentation, demand forecasting, churn prediction, and Apriori-based association rule mining.

MediSynth Insight – Healthcare Analytics Dashboard , *(Python, Pandas, Plotly Dash, Scikit-learn)*

- Designed and developed a data processing pipeline to generate, clean, and transform a 10,000-record synthetic healthcare dataset.
- Implemented preprocessing techniques, including missing value imputation, outlier detection and handling, and Min-Max normalization.
- Performed feature engineering, statistical analysis, and data aggregation using Pandas and Scikit-learn for downstream analytics.
- Developed interactive Plotly Dash dashboards with dynamic filtering, KPI tracking, and visualizations to uncover healthcare trends.

Ordinus - Intelligent Academic Timetable Generator , *(FastAPI, PostgreSQL, React)*

- Designed structured relational data models for faculty availability, course allocation, laboratory scheduling, and timetable constraints.
- Developed a constraint-based scheduling engine using backtracking and genetic algorithms to process scheduling data and generate optimized timetables.
- Implemented conflict detection and validation mechanisms to ensure consistency across multiple classes and faculty assignments.
- Built backend APIs and PostgreSQL integrations to manage scheduling data and support timetable generation.

Roles

Class Representative
06/2026 – Present

Student Internship Coordinator, CIT
2024 – 2025

Treasurer, FOSS CLUB, CIT
2024 – 2025

Event Manager, FOSS CLUB, CIT
2023 – 2024

Co-Curricular Activities

Achievements

- Winner - VisionX Hackathon - VISTA Engg solutions
- Winner -Codevortex - Computing department, CIT
- 2nd Runner up - Frontend Hackathon - FOSS-CIT

Technical Speaker & Mentor, FOSS-CIT

- Conducted Data Structures using C workshops (Coding Season 2 & 3), mentoring students through hands-on coding.
- Delivered Figma/UI-UX Design (Figma Unleashed) sessions, teaching design principles and interactive prototyping.

Certificates

Microsoft Research — Large Language Models and Their Applications
Cybersecurity - CISCO Networking Academy — Introduction to Cybersecurity
Figma - Udemy — Learn Figma in 14 days : Master UI Design & Prototyping